Intermediate Value Theorem

An End-to-End Meta-Math Walkthrough

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Abstract

A complete authored walkthrough of intermediate value theorem, with explicit mathematical contracts, local verification gates, and reusable exports. The report presents the target theorem, declared imports, a dependency graph, five-faculty node contracts, a continuous proof, a worked example, and a hypothesis-repair audit. It is an authored mathematical exposition, not a deployed-agent transcript or a formal proof-assistant certificate.

RA-01 / Real Analysis / Report v1

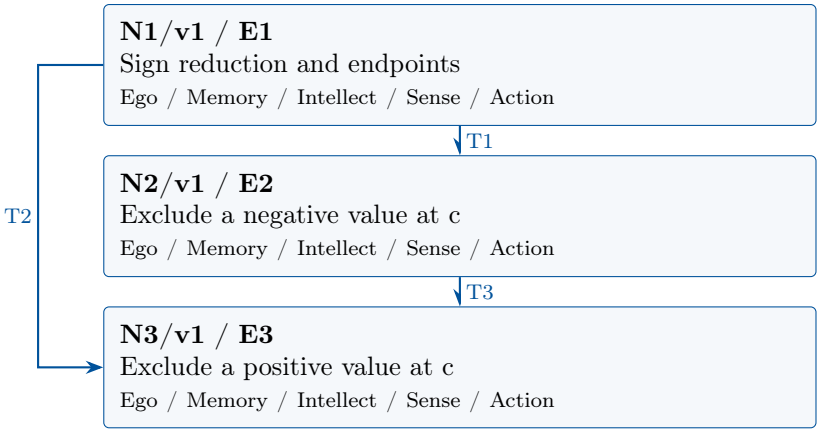
I. TARGET AND IMPORT CONTRACT

Theorem I.1. *For continuous $f : [a, b] \to \mathbb{R}$, $a < b$, prove that every y between $f(a)$ and $f(b)$ equals $f(c)$ for some $c \in [a, b]$.*

A. Foundations and permitted imports

Use the least-upper-bound property of R and epsilon–delta continuity. A continuous function at c preserves a strict sign on a sufficiently small relative neighborhood. No intermediate-value or root-existence theorem is imported.

II. NODE INTERACTION WORKFLOW



Each box is one mathematical agent containing all five faculties. Blue arrows carry the versioned exports listed below. The layout gives a reading order; only drawn arrows assert dependencies. Diagnostic repair is separate from the proof DAG.

A. Versioned evidence-transfer ledger

T1: N1/E1/v1 to N2. A supremum candidate c for the strict-sign case.

Receiving activity. N2: Exclude a negative value at c

Compatibility. Use the same mathematical objects and hypotheses as the source export. Receiving contract: Supremum candidate; continuity; prior exports E1/v1

T2: N1/E1/v1 to N3. A supremum candidate c for the strict-sign case.

Receiving activity. N3: Exclude a positive value at c

Compatibility. Use the same mathematical objects and hypotheses as the source export. Receiving contract: Supremum and nonempty strict-negative neighborhood of a; prior exports E1/v1, E2/v1

T3: N2/E2/v1 to N3. $c < b$ and $g(c) \geq 0$.

Receiving activity. N3: Exclude a positive value at c

Compatibility. Use the same mathematical objects and hypotheses as the source export. Receiving contract: Supremum and nonempty strict-negative neighborhood of a; prior exports E1/v1, E2/v1

III. AGENT ROLES AND FIVE-FACULTY ACTIVITIES

A. N1/v1: Sign reduction and endpoints

Object and role. Translate and possibly reverse the sign.

Inputs. Continuous f on[a,b]; y between endpoint values; foundational imports stated in §1

E — Ego

Own E1/v1: A supremum candidate c for the strict-sign case.

M — Memory

Retain Continuous f on[a,b]; y between endpoint values; foundational imports stated in §1; keep their assumptions and versions attached.

I — Intellect

Translate and possibly reverse the sign.

S — Sense

Endpoint equalities must be closed before assuming strict signs.

A — Action

Carry out the numbered deductions and publish E1/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N1.1 If $y = f(a)$ or $f(b)$, the corresponding endpoint solves the problem.

N1.2 Otherwise put $g = f - y$ and, if necessary, replace g by $-g$ so $g(a) < 0 < g(b)$.

N1.3 Define $S = \{x \in [a, b] : g(x) < 0\}$.

N1.4 It contains a and is bounded above by b, hence $c = \sup S$ exists.

Export E1/v1. A supremum candidate c for the strict-sign case.

Acceptance gate. Endpoint equalities must be closed before assuming strict signs. Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N1: Endpoint equalities must be closed before assuming strict signs. Recheck dependent nodes N2, N3.

B. N2/v1: Exclude a negative value at c

Object and role. A negative value would allow a larger point in S.

Inputs. Supremum candidate; continuity; prior exports E1/v1

E — Ego

Own E2/v1: $c < b$ and $g(c) \geq 0$.

M — Memory

Retain Supremum candidate; continuity; prior exports E1/v1; keep their assumptions and versions attached.

I — Intellect

A negative value would allow a larger point in S.

S — Sense

Use the endpoint sign to ensure there is room to the right.

A — Action

Carry out the numbered deductions and publish E2/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N2.1 Continuity at b and $g(b) > 0$ give a relative neighborhood of b with positive g, so $c < b$.

N2.2 If $g(c) < 0$, continuity gives $\delta > 0$ with g negative near c.

N2.3 Choose x with $c < x < \min(b, c + \delta)$, so $x \in S$ contradicting the upper-bound property.

N2.4 Thus $g(c) \geq 0$.

Export E2/v1. $c < b$ and $g(c) \geq 0$.

Acceptance gate. Use the endpoint sign to ensure there is room to the right. Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N2: Use the endpoint sign to ensure there is room to the right. Recheck dependent nodes N3.

C. N3/v1: Exclude a positive value at c

Object and role. Approximate the supremum from below.

Inputs. Supremum and nonempty strict-negative neighborhood of a; prior exports E1/v1, E2/v1

E — Ego

Own E3/v1: Every intermediate endpoint value is attained.

M — Memory

Retain Supremum and nonempty strict-negative neighborhood of a; prior exports E1/v1, E2/v1; keep their assumptions and versions attached.

I — Intellect

Approximate the supremum from below.

S — Sense

Being a supremum alone does not imply belonging to S.

A — Action

Carry out the numbered deductions and publish E3/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N3.1 Continuity at a gives some points strictly greater than a in S, so $c > a$.

N3.2 If $g(c) > 0$, g is positive on $(c - \delta, c + \delta) \cap [a, b]$.

N3.3 By the defining property of supremum there is $s \in S$ with $c - \delta < s \leq c$, a contradiction.

N3.4 Hence $g(c) \leq 0$.

N3.5 Together the inequalities force $g(c) = 0$; undoing the sign reduction gives $f(c) = y$.

Export E3/v1. Every intermediate endpoint value is attained.

Acceptance gate. Being a supremum alone does not imply belonging to S. Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N3: Being a supremum alone does not imply belonging to S. Withhold this terminal result until its gate passes.

IV. CONTINUOUS MATHEMATICAL PROOF

Read these steps in order as one proof; the diagram is optional. Numbered activities identify the specific constructions and deductions owned by each node.

A. §4.N1 — Sign reduction and endpoints

1. If $y = f(a)$ or $f(b)$, the corresponding endpoint solves the problem.
2. Otherwise put $g = f - y$ and, if necessary, replace g by $-g$ so $g(a) < 0 < g(b)$.
3. Define $S = \{x \in [a, b] : g(x) < 0\}$.
4. It contains a and is bounded above by b, hence $c = \sup S$ exists.

B. §4.N2 — Exclude a negative value at c

1. Continuity at b and $g(b) > 0$ give a relative neighborhood of b with positive g, so $c < b$.
2. If $g(c) < 0$, continuity gives $\delta > 0$ with g negative near c.
3. Choose x with $c < x < \min(b, c + \delta)$, so $x \in S$ contradicting the upper-bound property.
4. Thus $g(c) \geq 0$.

C. §4.N3 — Exclude a positive value at c

1. Continuity at a gives some points strictly greater than a in S, so $c > a$.
2. If $g(c) > 0$, g is positive on $(c - \delta, c + \delta) \cap [a, b]$.
3. By the defining property of supremum there is $s \in S$ with $c - \delta < s \leq c$, a contradiction.
4. Hence $g(c) \leq 0$.
5. Together the inequalities force $g(c) = 0$; undoing the sign reduction gives $f(c) = y$.

□

V. WORKED ORCHESTRATION

For $f(x) = x^3 - x - 1$, $f(1) = -1$ and $f(2) = 5$. The supremum construction yields a root in (1,2). Bisection checks $f(1.25) = -0.296875$ and $f(1.5) = 0.875$, narrowing the interval; these computations illustrate the theorem but are not its universal proof.

A. Boundary and hypothesis audit

The jump $g(x) = -1$ for $x < 0$ and 1 for $x \geq 0$ $on[-1, 1]$ has both signs and misses0. At its supremum candidate continuity fails. Sense rejects the sign-preservation step; no bracket computation repairs the original discontinuous theorem.

VI. ILLUSTRATIVE FAILURE AND REPAIR

The following is an authored diagnostic scenario. Changed assumptions define a new contract; they do not retroactively prove the original statement.

A. Failure and correction

This is an authored teaching trace, not a recorded agent run.

1. **Ego → Memory:** register the §1 theorem, imports, and requested exports before beginning the proof.
2. **Intellect → Action:** choose the bridges in §3 and carry out the numbered activities in §4; transmit only established E-exports along the §2 DAG.
3. **Sense → responsible node:** test the specific boundary/failure in §5 and the node's individual audit. A failing candidate is withheld; downstream nodes may not use it.
4. **Memory + Intellect:** identify whether the error is an omitted condition, an invalid inference, or an attempted extension beyond the theorem. Restore the exact contract or explicitly state a different one.
5. **Action:** redo the affected mathematical activity using the repaired condition or inference. Recheck consuming nodes, rather than feeding their conclusions back as premises.
6. **Ego:** publish only the supported export with its assumptions and proof location. A hypothetical revised export is v2; the written proof here is the v1 exposition, not a claim that this repair was executed.

B. Hypothesis boundary

The jump $g(x) = -1$ for $x < 0$ and 1 for $x \geq 0$ $on[-1, 1]$ has both signs and misses0. At its supremum candidate continuity fails. Sense rejects the sign-preservation step; no bracket computation repairs the original discontinuous theorem.

C. Version and recheck

Treat any changed assumption as a new contract. Recheck the revised export and all of its descendants. A backward diagnostic never establishes an earlier theorem. This is an authored teaching trace, not an execution log.

Provenance. This authored proof is complete relative to its declared imports. Reuse each export with its assumptions, proof, and version. The original Markdown remains available for comparison.